Friday, Aug 26

Sampling Distributions — Continued

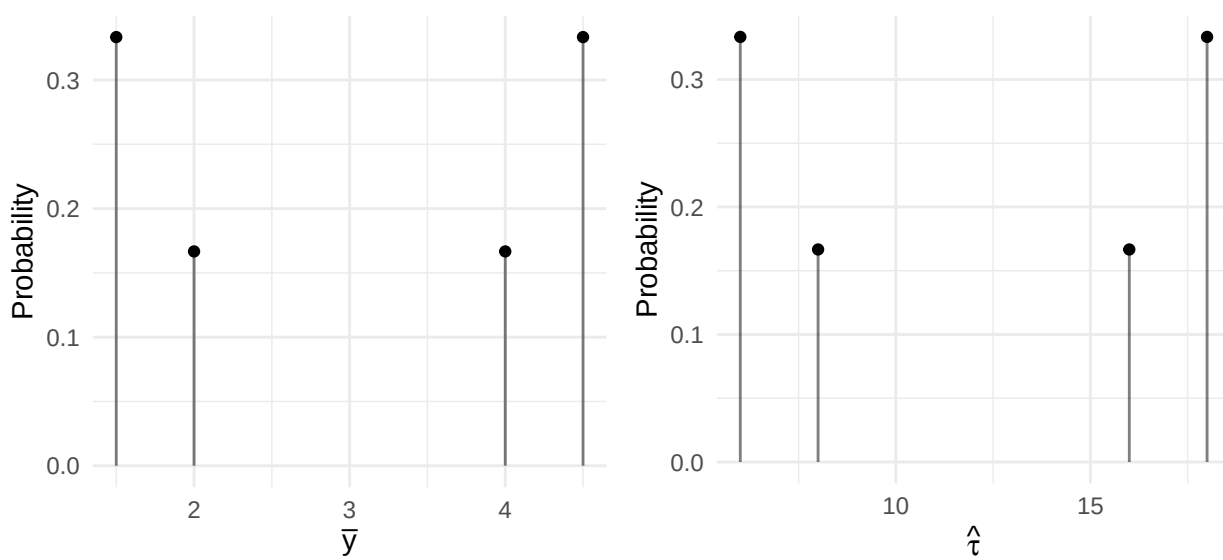
Example: Suppose we have a population of $N = 4$ elements where the target variable values are 1, 2, 2, and 7. The table below shows the **sample space** for a simple random sampling design with $n = 2$ as well as the values of the estimators. $\hat{\tau}$ and $\bar{y}$ for each sample.

Sample	$\hat{\tau}$	$\bar{y}$
$\{\mathcal{E}_1, \mathcal{E}_2\}$	6	1.5
$\{\mathcal{E}_1, \mathcal{E}_3\}$	6	1.5
$\{\mathcal{E}_1, \mathcal{E}_4\}$	16	4.0
$\{\mathcal{E}_2, \mathcal{E}_3\}$	8	2.0
$\{\mathcal{E}_2, \mathcal{E}_4\}$	18	4.5
$\{\mathcal{E}_3, \mathcal{E}_4\}$	18	4.5

Note that a **sample space** is simply the set of all possible samples. Here there are 6 samples in the sample space. The table below shows the *sampling distributions* of $\bar{y}$ and $\hat{\tau}$.

$\bar{y}$	$\hat{\tau}$	Probability
1.5	6	1/3
2.0	8	1/6
4.0	16	1/6
4.5	18	1/3

The figures below show the sampling distributions of $\bar{y}$ and $\hat{\tau}$.



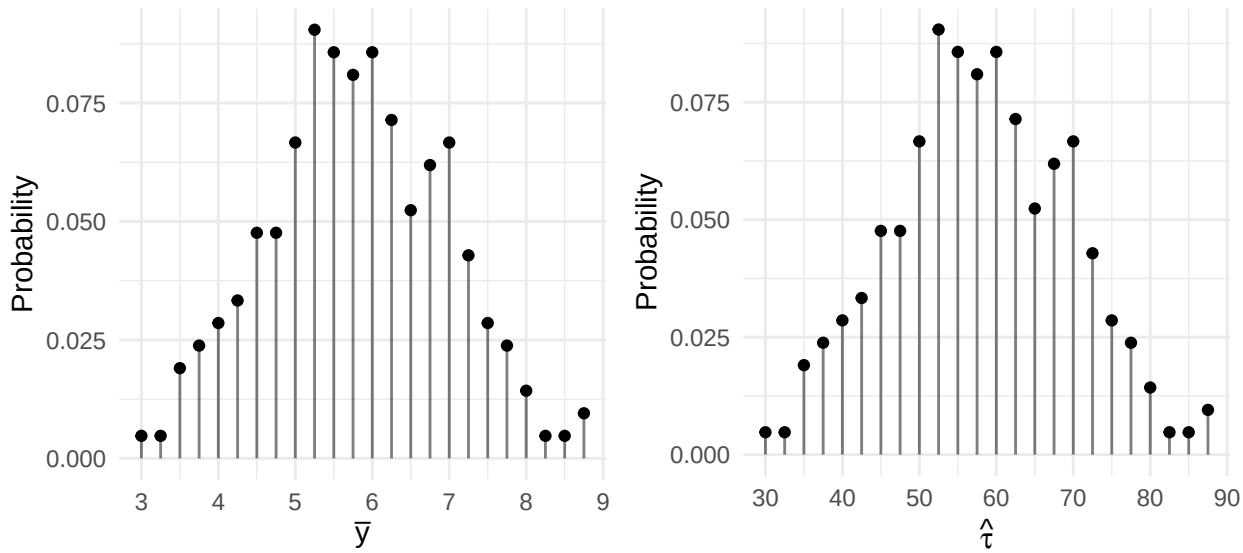
Example: Suppose now we have a population of $N = 10$ elements where the target variable values are 2, 3, 3, 4, 5, 6, 6, 9, 10, and 10. The table below shows the (abbreviated) **sample space** for a simple random sampling design with $n = 4$ as well as the values of the estimators $\hat{\tau}$ and $\bar{y}$ for each sample.

Sample	$\hat{\tau}$	$\bar{y}$
$\{\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3, \mathcal{E}_4\}$	30	3
$\{\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3, \mathcal{E}_5\}$	32.5	3.25
$\{\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3, \mathcal{E}_6\}$	35	3.5
$\{\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3, \mathcal{E}_7\}$	35	3.5
$\{\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3, \mathcal{E}_8\}$	42.5	4.25
$\{\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3, \mathcal{E}_9\}$	45	4.5
$\{\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3, \mathcal{E}_{10}\}$	45	4.5
$\{\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_4, \mathcal{E}_5\}$	35	3.5
$\vdots$	$\vdots$	$\vdots$
$\{\mathcal{E}_7, \mathcal{E}_8, \mathcal{E}_9, \mathcal{E}_{10}\}$	87.5	8.75

Here there are 210 samples in the sample space. The table below shows the *sampling distributions* of $\bar{y}$ and $\hat{\tau}$.

$\bar{y}$	$\hat{\tau}$	Probability
3.00	30.0	0.0047619
3.25	32.5	0.0047619
3.50	35.0	0.0190476
3.75	37.5	0.0238095
4.00	40.0	0.0285714
4.25	42.5	0.0333333
4.50	45.0	0.0476190
4.75	47.5	0.0476190
5.00	50.0	0.0666667
5.25	52.5	0.0904762
5.50	55.0	0.0857143
5.75	57.5	0.0809524
6.00	60.0	0.0857143
6.25	62.5	0.0714286
6.50	65.0	0.0523810
6.75	67.5	0.0619048
7.00	70.0	0.0666667
7.25	72.5	0.0428571
7.50	75.0	0.0285714
7.75	77.5	0.0238095
8.00	80.0	0.0142857
8.25	82.5	0.0047619
8.50	85.0	0.0047619
8.75	87.5	0.0095238

The figures below show the sampling distributions of $\bar{y}$ and $\hat{\tau}$.



Means and Variances of Random Variables

If we have a *discrete* random variable X, then the **mean** of that random variable (also called its *expectation* or *expected value*) is

$$E(X) = \sum_x xP(x),$$

where the x below $\sum$ indicates that we sum over all values of x.

Example: We can easily confirm that based on the sampling distributions for the smaller population above that $E(\bar{y}) = 3$ and $E(\hat{\tau}) = 12$.

But there is a shortcut if we use simple random sampling. For *any* simple random sampling design, *it can be shown that* $E(\bar{y}) = \mu$ and $E(\hat{\tau}) = \tau$. When the mean of an estimator equals what is being estimated, then we say that the estimator is an **unbiased** estimator (otherwise the estimator is a **biased** estimator).

Warning: We have discussed *two* distinct means here: the population mean (μ), and the mean of an estimator — e.g., $E(\bar{y})$ and $E(\hat{\tau})$. Don't confuse them!

If we have a discrete random variable X, then the **variance** of that random variable is

$$\mathrm{V}(X) = \sum_x [x - E(X)]^2 P(x).$$

Example: We can easily confirm that based on the sampling distributions for the smaller population above that $\mathrm{V}(\bar{y}) \approx 1.83$ and $\mathrm{V}(\hat{\tau}) \approx 29.33$.

It can be shown that *under simple random sampling* that the variances can be computed as

$$\mathrm{V}(\bar{y}) = \left(1 - \frac{n}{N}\right)\frac{\sigma^2}{n} \quad \text{and} \quad \mathrm{V}(\hat{\tau}) = N^2\left(1 - \frac{n}{N}\right)\frac{\sigma^2}{n},$$

where σ^2 is the *population variance*

$$\sigma^2 = \frac{1}{N-1}\sum_{i=1}^{N}(y_i - \mu)^2,$$

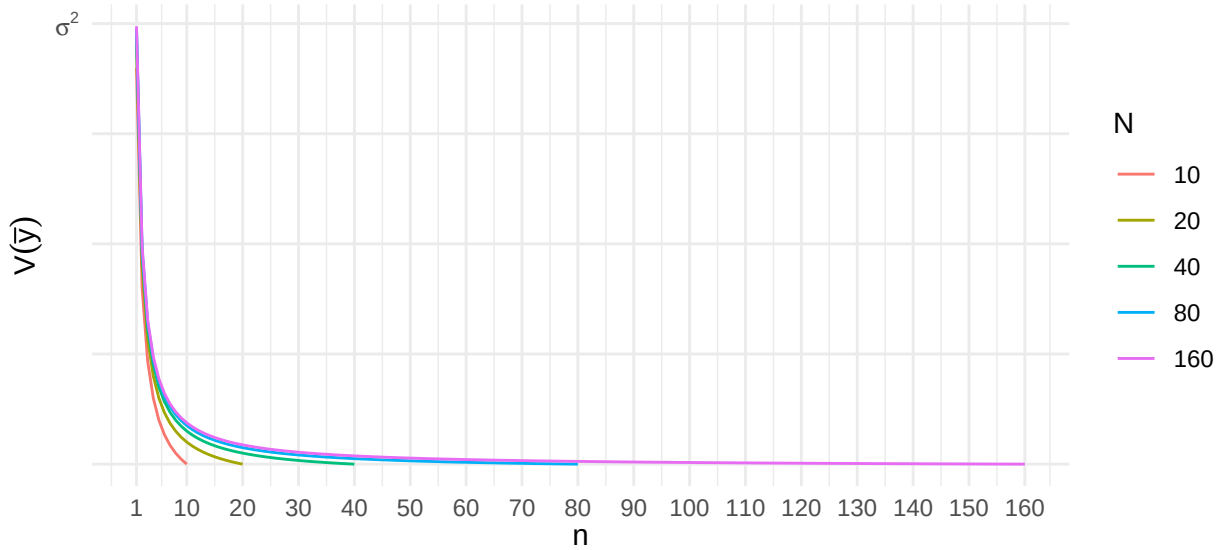
and where μ is the population mean.[1]

Example: For the smaller population above $\sigma^2 = 7.33$, and for the larger population $\sigma^2 \approx 8.84$. What are $\mathrm{V}(\bar{y})$ and $\mathrm{V}(\hat{\tau})$ for each design?

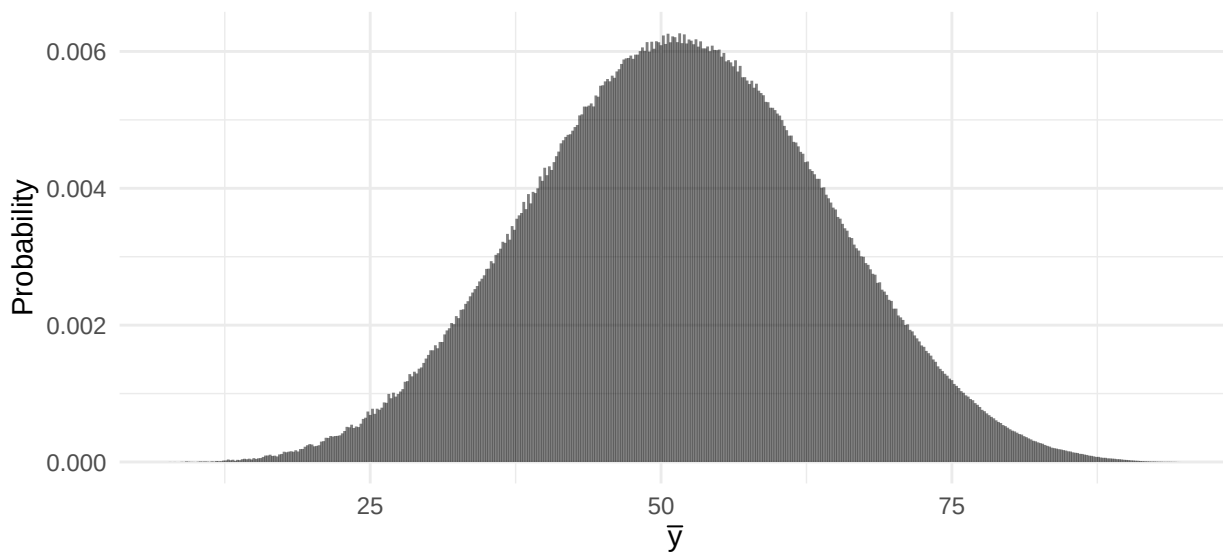
Warning: We've discussed *two* distinct variances here: the *population variance* (σ^2) and the *variance of an estimator* — e.g., $\mathrm{V}(\bar{y})$ and $\mathrm{V}(\hat{\tau})$. Don't confuse them!

[1]Some textbooks will define σ^2 by dividing by N instead of $N-1$. This can be done but it changes the formulas for $V(\bar{y})$ and $V(\hat{\tau})$ in a way that I find awkward for later developments. In terms of our interpretation of σ^2 this is inconsequential since we rarely care about σ^2 in isolation and the difference between N and $N-1$ is very small when N is large which is frequently the case in survey sampling.

How do n and N affect the variance of these estimators?



Central Limit Theorem

What can we say about the *shape* of the sampling distribution of $\bar{y}$ or $\hat{\tau}$? The **central limit theorem** for simple random sampling states that as n, N, and $N - n$ increase, and assuming some other rather technical but usually applicable conditions, the sampling distribution of $\bar{y}$ "approaches" a normal distribution. This implies the same behavior for the sampling distribution of $\hat{\tau}$.

Example: Consider simple random sampling design with a population of $N = 50$ elements and a sample size of $n = 5$. The values of the target variable in the population were selected randomly from the integers from 1 and 100. The figure below shows the sampling distribution of $\bar{y}$.